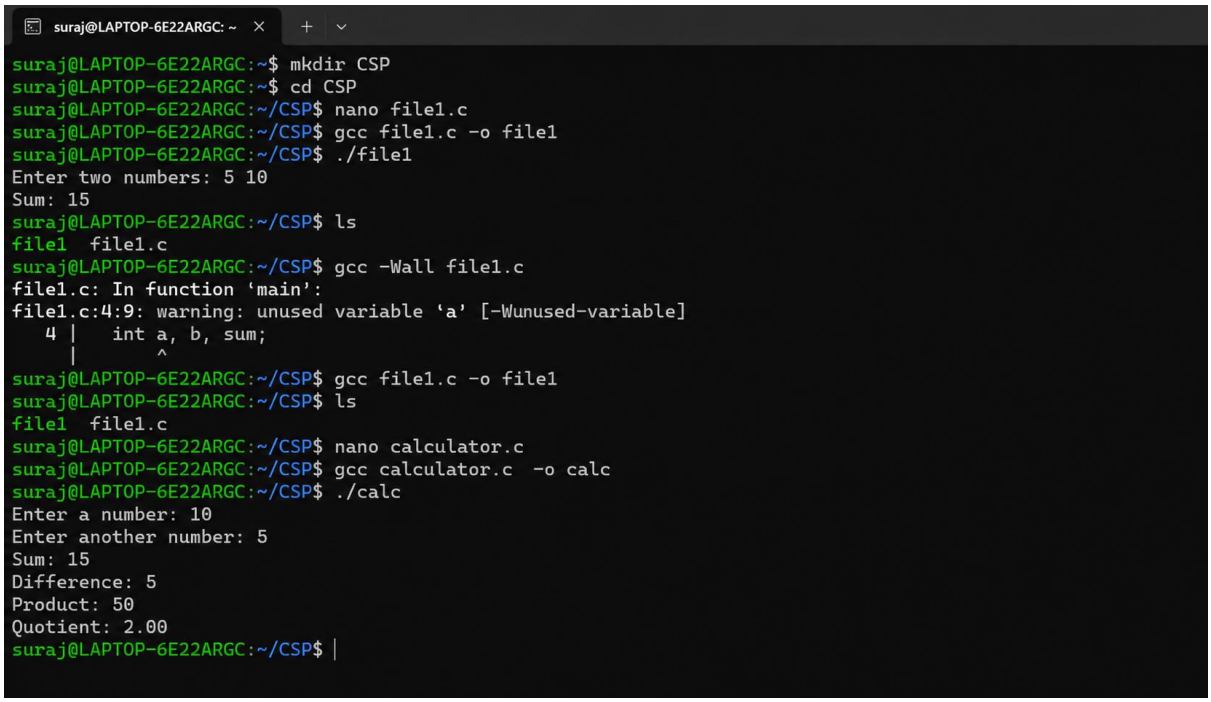
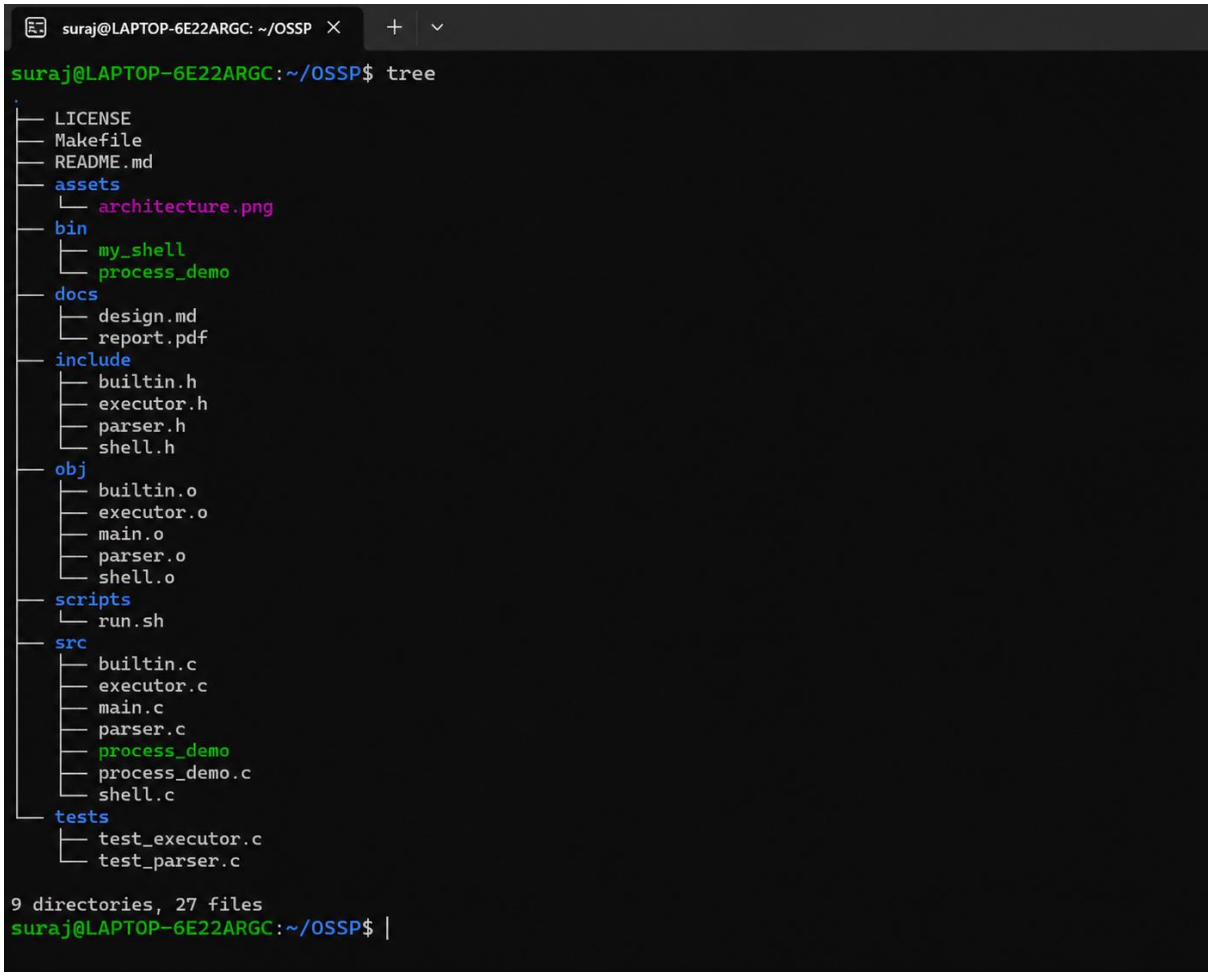
SKILL- Week 2

1.) Assignment-1



```
suraj@LAPTOP-6E22ARGC:~$ mkdir CSP
suraj@LAPTOP-6E22ARGC:~$ cd CSP
suraj@LAPTOP-6E22ARGC:~/CSP$ nano file1.c
suraj@LAPTOP-6E22ARGC:~/CSP$ gcc file1.c -o file1
suraj@LAPTOP-6E22ARGC:~/CSP$ ./file1
Enter two numbers: 5 10
Sum: 15
suraj@LAPTOP-6E22ARGC:~/CSP$ ls
file1  file1.c
suraj@LAPTOP-6E22ARGC:~/CSP$ gcc -Wall file1.c
file1.c: In function 'main':
file1.c:4:9: warning: unused variable 'a' [-Wunused-variable]
   4 |   int a, b, sum;
     |       ^
suraj@LAPTOP-6E22ARGC:~/CSP$ gcc file1.c -o file1
suraj@LAPTOP-6E22ARGC:~/CSP$ ls
file1  file1.c
suraj@LAPTOP-6E22ARGC:~/CSP$ nano calculator.c
suraj@LAPTOP-6E22ARGC:~/CSP$ gcc calculator.c  -o calc
suraj@LAPTOP-6E22ARGC:~/CSP$ ./calc
Enter a number: 10
Enter another number: 5
Sum: 15
Difference: 5
Product: 50
Quotient: 2.00
suraj@LAPTOP-6E22ARGC:~/CSP$
```



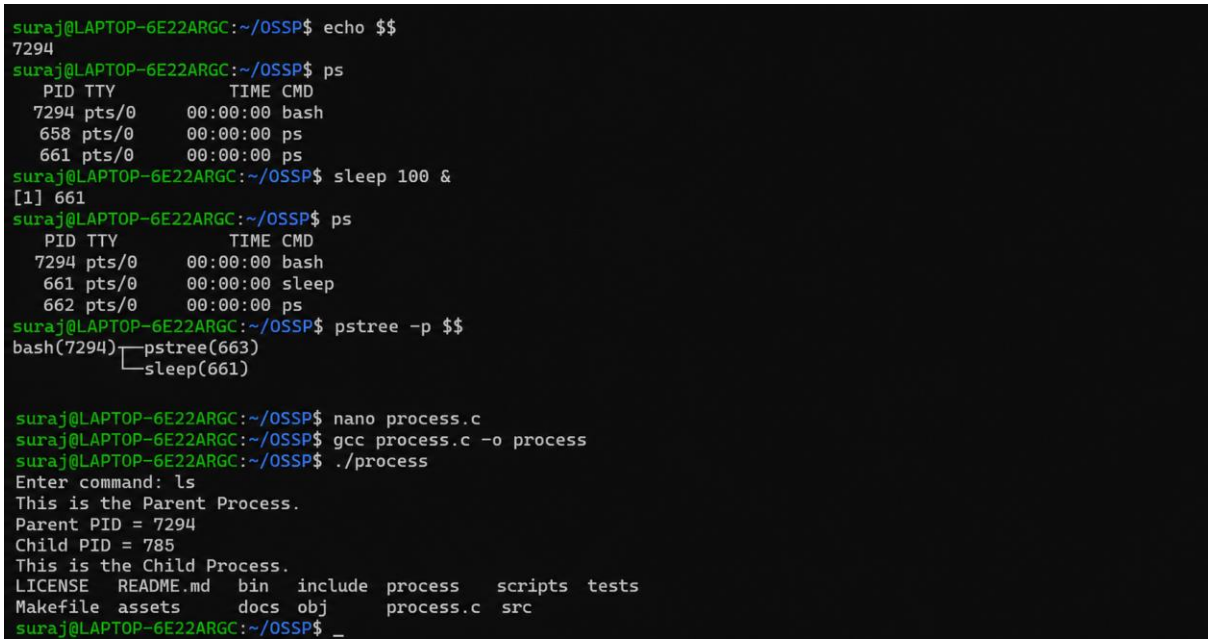
```
suraj@LAPTOP-6E22ARGC:~/OSSP$ tree
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
│   ├── my_shell
│   └── process_demo
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── obj
│   ├── builtin.o
│   ├── executor.o
│   ├── main.o
│   ├── parser.o
│   └── shell.o
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   ├── process_demo
│   ├── process_demo.c
│   └── shell.c
└── tests
    ├── test_executor.c
    └── test_parser.c

9 directories, 27 files
suraj@LAPTOP-6E22ARGC:~/OSSP$
```

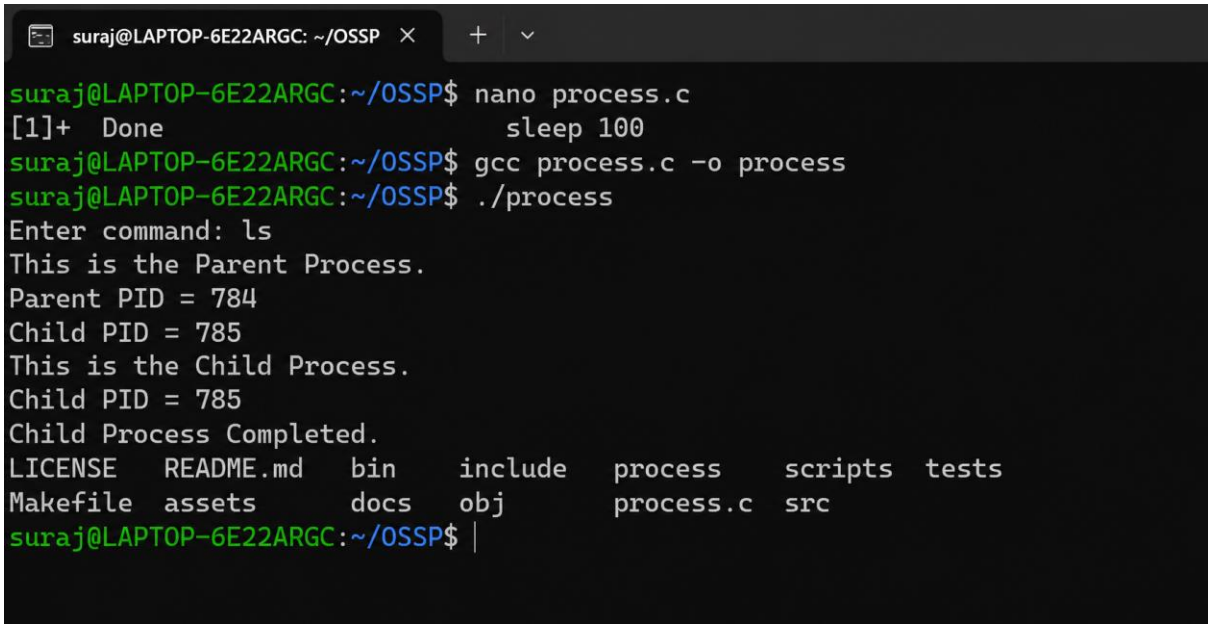
This assignment creates an organized OSSP project structure containing source files, header files, tests, documentation and scripts. mkdir is used to create directories, touch creates files, and tree displays the complete project structure in tree format.

2.) Assignment-2

Task 1:-



Task 2 :-



fork() creates a child process from the parent process and gives both processes separate process IDs. exec() allows the child process to execute the command entered by the user while the parent waits for the child to complete.